

Technical Document #645: Computer Vision - Comprehensive Overview

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Optical character recognition (OCR) converts images of text into machine-readable text. It is used in document digitization and automation.

Image classification assigns a label to an image from a fixed set of categories. CNNs like ResNet and EfficientNet achieve superhuman performance on benchmark datasets.

Pose estimation detects human body keypoints in images. It is used in action recognition, animation, and sports analysis.

Visual question answering (VQA) answers questions about images. It requires both visual understanding and language comprehension.

Video understanding analyzes temporal dynamics in video sequences. 3D CNNs and transformer-based models capture spatiotemporal features.

Semantic segmentation classifies each pixel in an image into a category. It is used in autonomous driving and medical image analysis.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Visual question answering (VQA) answers questions about images.
- Video understanding analyzes temporal dynamics in video sequences.
- Image generation creates new images from random noise or text descriptions.